COMP7993

Early Deliverable / Critical Review Assessment

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Programme: MSc in Artificial Intelligence

Paper Reference:

Jackins, V., Vimal, S., Kaliappan, M., & Lee, M. Y. (2021). *AI-based smart prediction of clinical disease using random forest classifier and Naive Bayes*. The Journal of Supercomputing, 77, 5198–5219.

Paper link: https://doi.org/10.1007/s11227-020-03481-x

Summary

Motivation

The paper *AI-based Smart Prediction of Clinical Disease Using Random Forest Classifier and Naive Bayes* addresses the increasing need for accurate and efficient disease prediction systems in modern healthcare. As the number of patients suffering from chronic diseases such as diabetes, coronary heart disease, and breast cancer continues to grow, there is a strong demand for intelligent tools that can assist healthcare professionals in making early diagnoses. Traditional diagnostic methods often require considerable time and resources, while machine learning techniques have the potential to analyse large amounts of patient data quickly and identify hidden patterns. The authors aim to investigate whether machine learning algorithms can improve disease prediction accuracy and support medical decision-making.

Contribution

The main contribution of the paper is the development and evaluation of an AI-based disease prediction framework using two supervised machine learning algorithms: Random Forest and Naive Bayes. The study demonstrates how these algorithms can be applied to multiple publicly available medical datasets to classify patients according to their likelihood of having a particular disease. Rather than focusing on a single illness, the research evaluates the performance of both algorithms across different clinical domains, providing a broader understanding of their applicability in healthcare. The work also offers a comparative analysis to determine which algorithm performs better under similar conditions.

Methodology

The authors collected publicly available datasets related to diabetes, coronary heart disease, and breast cancer and applied standard preprocessing techniques to prepare the data for analysis. After cleaning and organising the datasets, they divided them into training and testing subsets. Both the Naive Bayes classifier and the Random Forest classifier were trained using the prepared data and evaluated using common performance measures such as classification accuracy and confusion matrices. The Naive Bayes algorithm estimates class probabilities based on conditional probability assumptions, whereas Random Forest constructs multiple decision trees and combines their outputs to improve predictive performance. The comparison between these two approaches allows the researchers to evaluate their effectiveness for medical diagnosis.

Conclusion

The experimental results indicate that both machine learning methods are capable of predicting diseases with good accuracy, making them suitable candidates for clinical decision support systems. However, the Random Forest classifier generally achieved higher predictive performance than Naive Bayes across the datasets considered in the study. Despite this, the authors highlight that Naive Bayes remains a computationally efficient and straightforward algorithm that can still provide useful predictions, particularly in situations where simplicity and execution speed are important. Overall, the paper concludes that artificial intelligence techniques can play an important role in assisting healthcare professionals and improving early disease detection through data-driven analysis.

Critique

The paper presents a well-structured comparison between the Random Forest and Naive Bayes algorithms for clinical disease prediction and demonstrates the usefulness of machine learning in healthcare. However, one limitation is that the evaluation is restricted to only two classification algorithms. Including additional methods such as Support Vector Machines, Logistic Regression, or Gradient Boosting could have provided a more comprehensive benchmark and allowed readers to better understand the relative strengths and weaknesses of the proposed approaches.

Another point of concern is the limited discussion of model interpretability. While the study reports prediction accuracy and classification results, it does not sufficiently explain why the algorithms arrive at specific predictions. In medical applications, healthcare professionals often require explanations for automated decisions before trusting and adopting such systems in practice. Providing feature importance analysis or explainable AI techniques would have improved the practical value of the research.

Finally, although the authors use multiple publicly available datasets, the study does not investigate how well the trained models would perform on external clinical data collected from different hospitals or populations. Medical datasets can vary significantly across regions and demographic groups, and testing on independent datasets would strengthen confidence in the robustness and generalisability of the proposed approach.

Synthesis

The research can be further extended by developing a web-based healthcare decision support system that integrates multiple Naive Bayes models trained on publicly available medical datasets. In addition to predicting diabetes, coronary heart disease, and breast cancer, the platform could also incorporate prediction models for hypertension, stroke risk, chronic kidney disease, and liver disease. This would allow users to assess their likelihood of developing a broader range of common health conditions through a single, user-friendly application.

The proposed system could be implemented using a Flask-based backend and an intuitive web interface where users enter commonly known medical and lifestyle information, such as age, gender, height, weight, body mass index (BMI), blood pressure, blood glucose level, cholesterol level, smoking status, alcohol consumption, physical activity, and family medical history. Based on these inputs, the application would generate and display ranked probability estimates for the supported diseases in real time, helping users better understand their potential health risks and encouraging them to seek professional medical advice when necessary.

Another possible extension would be to incorporate explainable artificial intelligence techniques and personalised health recommendations. Alongside displaying the predicted probabilities, the application could highlight the key factors influencing each prediction and provide guidance encouraging users to consult healthcare professionals when elevated risks are detected. Evaluating the system on larger and more diverse datasets from multiple populations would further improve its reliability and practical value for healthcare applications.